R68: fourth Frobenius lattice step

All eight classes of $\frac{V_3}{O}$ are squared with the actual degree-32 source multiplication table. Complete denominator clearing gives 25 binary remainder rows of rank four, hence four square survivors. Every remainder is reconstructed, every kernel product vanishes, and an altered kernel is rejected.

Fourth complete descending Frobenius step

Exact algebraic dimensions; run fourth-frobenius68-01. No timing comparison.

The full fourth PID lattice has rank 32. The quotient $\frac{V_4}{O}$ has binary dimension four and $\frac{V_3}{V_4}$ has dimension four. Both module inclusions, both field-basis inverse products, old-order inclusion, and all 32 new basis squares back in V_3 pass. An altered inverse is rejected.

Execution and evidence

One native attempt finished with exit 0, the exact success marker, and no Singular error lines. Its UTC wall interval is 176 s, a receipt rather than a controlled performance comparison. Six historical source byte pins and six donor hashes are checked in results.json. The complete quotient, binary kernel, and polynomial/field lattice bundles are saved.

This is the fourth descending step. The next calculation tests all four classes of $\frac{V_4}{O}$ and continues until the lattice stabilizes. Actual proper branches and divisors follow that affine normalization.

The method follows Singh–Swanson, Theorem 1.1.